

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

– TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of VPA Implementer (mandatory)

Appendix 3 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11188
Title of Project	GS7591 VPA 45 Murang'a Improved Cookstove Project (GS11188)
Time of First Submission Date	27/05/2021
Date of Design Certification	N/A
Version number of the VPA-DD	5
Completion date of version	25/10/2021
Coordinating/managing entity	Thomas Devesa – CO2balance UK Ltd Katri Hastings – CO2balance UK Ltd
VPA Implementer (s)	Wisdom Energy Hub Ltd
Project Participants and any communities involved	CO2balance UK Ltd
Host Country (ies)	Republic of Kenya
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	GS TPDDTEC v3.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☒ Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in Error! Reference source not found.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emissions Reductions	23,863	Emissions Reductions
3 Good Health and Well-Being	Total reduction in household air pollution for project activity in year y (%)	59%	HAPR _y
5 Gender Equality	Total reduction time spent collecting firewood and cooking for project activity in year y (%)	16%	TR _y
7 Affordable and Clean Energy	The number of active ICS disseminated by the project, during year y	2,400	ICS _{a,y}

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

CO2balance UK Ltd (CO2balance) and Wisdom Energy Hub will distribute around 3,000 local designed and manufactured stoves per year across Murang'a, Nyeri, Kirinyaga, Embu and Kiambu Counties in the Aberdare Highland region of Kenya, where households often cook on three stone fires or Kunu Mbili Improved Cookstoves (ICS). The project stoves will generally be sold at local markets and via agents, also known as Last Mile Distributors (LMDs).

The Aberdare Highland region is characterized by low penetration of clean cooking solutions and severe shortages of wood fuel. The main economic activities in the region are tea plantations. Local people can sell wood and timber to tea factories where they receive preferential prices. In addition, a Kenyan Government ban has disrupted the supply of wood fuel from the Aberdare Forest resulting in even higher costs in purchasing fuelwood.

The purpose of the Project is to reduce greenhouse gas emissions at a household level through the distribution of improved cookstoves which reduce wood consumption. As well as reducing greenhouse gas emissions, the Project will provide various benefits to the end user households and communities whilst helping to meet the UN Sustainable Development Goals (SDG) targets. Benefits include improved health through reduction in indoor air pollution, saving women time on unpaid chores, economic development by creating manufacturing jobs and sales opportunities, and reducing the income expenditure on fuelwood for stove recipients.

Carbon finance will be utilized to provide a subsidy on the Improved Cookstove developed by Wisdom Energy Hub Ltd, enabling greater penetration into the project area.

A.1.1. Eligibility of the project under approved PoA

This Project is automatically eligible for Gold Standard Certification as it follows the approved Gold Standard Activity Requirements 1.2(b) End-Use Energy Efficiency: efficient cooking, and eligible under the PoA as it is in one of the defined host

countries (section A.3), Republic of Kenya, and implements one of the defined project technologies (section A.4), improved cookstoves.

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	Types of Project	Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility Principles and Requirements section.	The projects involve the sale of locally designed and manufactured improved cookstoves. The project type is eligible under Community Services Activity Requirements 3.1.1(b) End-Use Energy Efficiency: efficient cooking.
2	Location of Project	Projects may be located in any part of the world.	The VPA is located in Kenya as specified in Section A.2 of this document. Kenya is one of the countries outlined in the PoA-DD. The VPA is uniquely defined by a range of GPS coordinates and current administrative maps to define the project boundary.
3	Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall	The project boundary, including GPS co-ordinates and maps are included in Section A.2 and B.3. This VPA is not included by any other carbon standard and will be capped at an energy output of 60Gwh (180GWhth) per year as per CDM small-scale requirements. The host country, Kenya, and location of each VPA is specified in Section A.2, in line with the locations outlined in Section A.3 of the PoA-DD. The project boundary, including GPS co-ordinates and maps are included in Sections A.2 and B.3.

		not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	This VPA is not included by any other carbon standard and will be capped at an energy output of 60Gwh (180GWhth) per year as per CDM small-scale requirements.
...	(d) Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The Project Activity does not violate any of the Host Country's legal, environmental, ecological and social regulations. The Project Activity <i>reduces</i> firewood consumption, which is inline with the Green Economy Strategy and Implementation Plan 2016 – 2030¹; and the economic empowerment of women through sales and monitoring through Womens' Groups is inline with the National Policy on Gender and Development².

¹ Kenya Green Economy Strategy and Implementation Plan 2016-2030, Ministry of Environment, Natural Resources, 2016. https://www.greengrowthknowledge.org/sites/default/files/downloads/policy-database/KENYA%29%20Improving%20Efficiency%20in%20Forestry%20Operations%20and%20Forest%20Product%20Processing%20in%20Keyna_0.pdf

² National Policy on Gender and Development, Sessional Paper No. 02 of 2019, The State Department of Gender Affairs. 2019. <https://gender.go.ke/wp-content/uploads/2021/04/Final-NPGAD-March-2021.pdf>

(e) Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Contact details of the Project Developer are included in Appendix 1.
(f) Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full	At the point of sale, purchasers of the stove are informed of the transfer of carbon rights in exchange for a stove subsidy. The terms of the carbon transfer are stated on a warranty card received at the time of purchase.

	and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	
(g) Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further Project implementation in affected areas.	There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity
(h) Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the OECD Development	A declaration confirming that there is no diversion of ODA has been submitted.

Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

In this project CO2balance, is the managing entity and is responsible for communication with the Gold Standard. CO2balance's implementing partner is Wisdom Energy Hub who manufacture the stoves and market them using Last Mile Distributors (LMDs) to sell the stoves. Wisdom/LMDs conduct the monitoring. CO2balance own the legal rights to the carbon credits created by the project.

The purchasers of the project stove are informed of the carbon transfer at the point of sale and receive a receipt which confirms the transfer of carbon rights. This entitles CO2balance to the rights of any carbon saved by the project.

A.2. Location of project

Murang'a, Nyeri, Kirinyaga, Embu and Kiambu Counties, Aberdare Highlands Region, Kenya



Figure 1. Project boundary within Republic of Kenya.

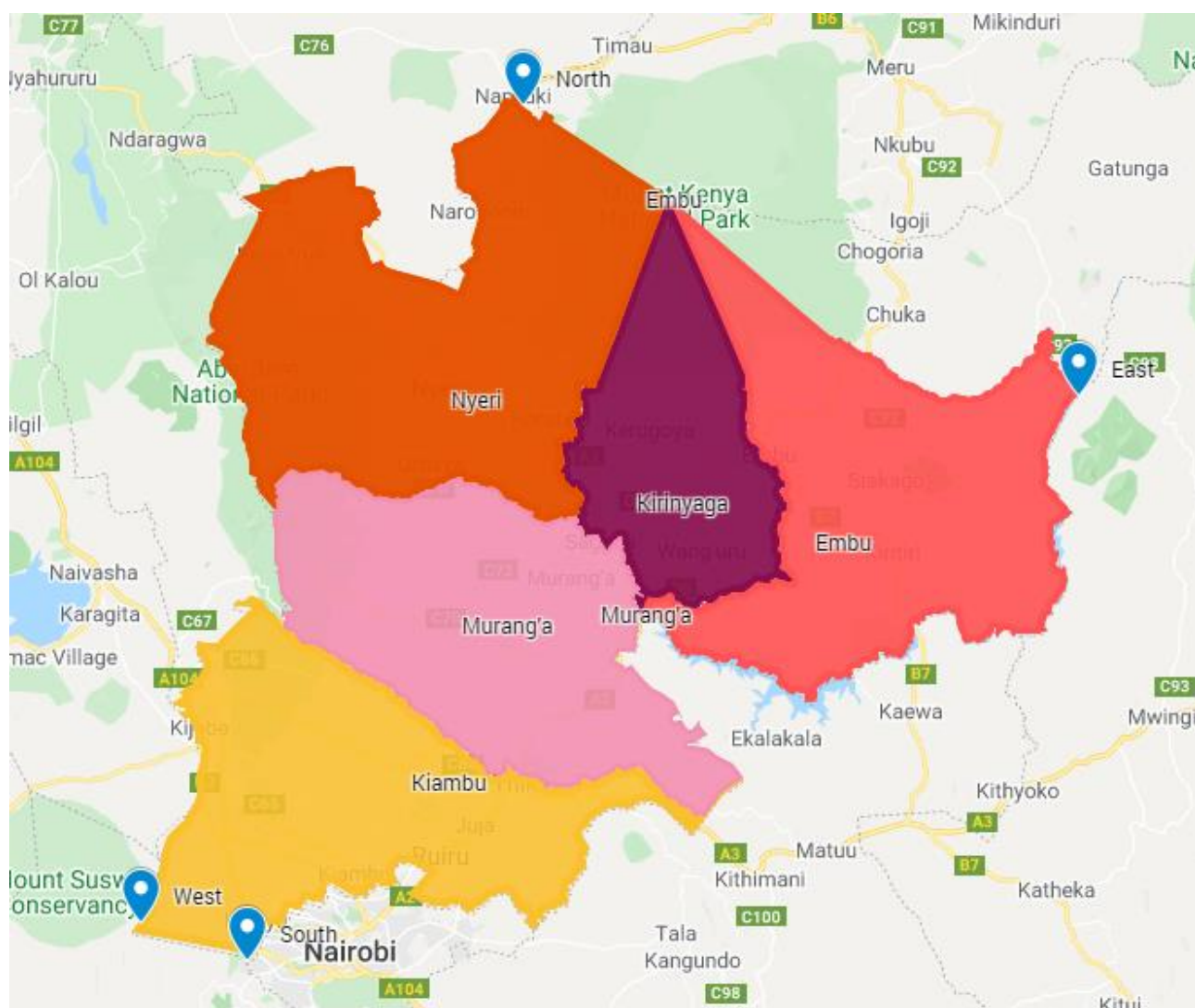


Figure 2. The project boundary in greater detail.

	Latitude	Longitude
North	-0.00494	37.08414
East	-0.44656	37.92981
South	-1.30658	36.66509
West	-1.24941	36.50239

A.3. Technologies and/or measures

Households, in the rural Highland areas of Murang'a, Nyeri, Kirinyaga, Embu and Kiambu Counties, often cook on traditional cook on traditional three stone fires and

Kunu Mbili ICS. Both technologies are inefficient and demand a lot of firewood, resulting in high CO₂ emissions, deforestation, and exposure to indoor air pollution, especially when compared to the project technology. The project stoves will displace the inefficient baseline technologies and reduce greenhouse gas emissions by reducing wood use by roughly 44%.

The reduced use and demand for firewood will reduce greenhouse gas emissions, reduce demand of wood collection on local forests (for those that collect wood), reduce requirement to purchase wood (for those who buy wood), reduce the time spent collecting wood and cooking, and reduce exposure to harmful indoor air pollution.

The project stoves are designed and manufactured locally by Wisdom Energy Hub Ltd. The stoves are sold at markets by Wisdom Stoves, who also employ Last Mile Distributors to distribute the stoves to the end users.



The Wisdom Jiko is an innovative natural draft gasifier stove featuring a galvanized steel cowling, mild steel fuel chamber and ceramic liner. It uses the TLUD (Top Lit Up Draft) technology. The stove has been modified through rigorous field tests to develop a design best able to serve rural households in Kenya, especially in the highlands. It is rated a tier 3 for ppm emissions and tier 2 for efficiency at 28%. All stoves are serialized by punching the number on the outer cowling for ease of trace back and management of warranty claims.

The stove primarily uses dry firewood as feed stock. Other fuels, such as non-carbonised briquettes, wood pellets and crop stems can also be used with the stove. The smoke from the feedstock is converted into a flammable gas known as synthetic or producer gas that burns with a clean, smokeless flame. The feedstock is then converted into quality bio-char which can continue burning in the stove for prolonged cooking. The bio-char can also be saved because it has additional uses especially in organic farming where it is utilized for soil augmentation. The new version dubbed Wisdom Jiko M3 retails for only KSH.3,900.

Each stove has a unique serial number to distinguish it from other stoves, when it is sold the householder is given a receipt identify it and to avoid double counting.

A.4. Scale of the project

This is a small scale “type II” project as it focuses on “energy efficiency improvement project activities” and must not exceed the equivalent of 60 gigawatt hours per year every year. VERs are capped at 60,000 per year.

A.5. Funding sources of project

The project is funded by CO2balance.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Gold Standard Methodology: Technologies and Practices that Displace Decentralized Thermal Energy Consumption (TPDDTEC) v3.1.

B.2. Applicability of methodology (ies)

The project is expressly applicable under section 1 of TPDDTEC v3.1.

Methodology Requirement	Project
1. 'The project boundary can be clearly identified, and the technologies counted in the project are not included in another voluntary market or CDM project activity.	The project area, the counties of Murang’a, Nyeri, Kirinyaga, Embu and Kiambu, have been clearly demarcated using political boundaries recognised in Kenya. Each technology will be individually tagged with an identification code which is stored securely in the project database. Regular project surveys together with distribution records will ensure that the technologies included in the project are not double counted.
2. Technologies have a continuous useful energy output of less than 150kW per unit (defined as total energy delivered usefully from start to end of operation of a unit divided by time of operation). For technologies or practices that do not	The estimated energy output of the project stove is 12.85 Kw which is well within the methodological limit of 150kw.

deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	
3. The use of the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology and the definitive discontinuity of its use.	As noted in the Gold Standard Methodology p.5. 'the removal and continued non-use of three stone fires and other easily constructed traditional devices (the baseline technology replaced by this project activity) is in many cases unlikely and impractical to monitor.' The project stove is purchased by users as they see some benefit in using it over their current woodfuel, such as saving on wood costs, saving time on wood collection, ability to move the stove, ease of use, and/or reduced smoke. Therefore, households are unlikely to use the baseline stoves if they have invested in the project stove.
a) The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology, whether the existing baseline technology is not surrendered at the time of the introduction of the improved technology, or whether a new baseline technology is acquired and put to	Overall use of the baseline technology will be monitored in conjunction with that of the project technology, as will the emergence of any other baseline technology by targeted end users. As per the Methodology kitchen surveys will be carried out at regular intervals to determine any changes in baseline technology use.

use by targeted end users during the project crediting period.	
b)'The success of the mechanism put into place must therefore be monitored, and the approach must be adjusted if proven unsuccessful. If an old technology remains in use in parallel with the improved technology, corresponding emissions must of course be accounted for as part of the project emissions.'	Parallel baseline technology use will be revealed during monitoring and its effect on emissions reductions will be captured in the KPT and usage surveys. The uptake rate U will also be determined by surveys and hence used to account for parallel baseline and project technology use.
4. The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. This must be communicated to the technology producers and the retailers of the improved technology or the renewable fuel in use in the project situation by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users should be notified that they cannot claim for emission reductions from the project.	Transfer of carbon rights is included in the receipt/warranty provided to stove recipients at the point of sale. A carbon copy is also kept by Wisdom Energy Hub. The LSC has been postponed due to Covid-19. Once an LSC is possible, the process of carbon rights transfer will be explained in detail.
5. Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as	As the technology used in this project has been specifically designed to displace baseline feedstock use via fuelwood, rather than a new biomass feedstock, this criterion is not applicable to this project. The emission reductions from this project will result from a change in

defined in the latest version of the Gold Standard rules.	quantity of fuel consumed, rather than change of fuel type.
a) Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases (as listed in section 2.1) emitted by the project fuel/stove combination are estimated with adequate precision. The project fuel/stove combination may include instances in which the project stove is a baseline stove.	The fuel used in both the project and baseline scenario is the same, that due to increase in efficiency of the stoves should be reduced by 59%
b) Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation, but may be used as data informing the equations in section 2.0 of this methodology if correlated to data on distribution and results of field tests and surveys confirming (a) actual use of the renewable fuel and usage patterns such as average fraction of non-renewable fuels used in mixed combustion or seasonal variation of fuel types, (b) GHG emissions, (c) evidence of CO levels not deteriorating (d) any further factors effecting emission reductions significantly.	Renewable fuels are not sold as part of this project therefore this point is not applicable.

B.3. Project boundary

Source		GHGs Included?		Justification/Explanation
Baseline	Combustion of firewood	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions

Project scenario		N ₂ O	Yes	Can be significant for some fuels
	Combustion of firewood	CO ₂	Yes	Important source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant for some fuels

B.4. Establishment and description of baseline scenario

The baseline scenario has been established inline with GS4GG *Principles & Requirements* and the TPDDTEC v3.1 methodology. The baseline study consisted of 111 Kitchen Surveys and 42 Kitchen Performance Tests. The KS contains 43 questions regarding demographics, household characteristics, kitchen habits, fuel use and health. Baseline surveying was conducted in highland and lowland areas, in part to best determine areas to focus project stove sales.

The baseline study covered 5 districts within Murang'a County. The baseline study sample was based on 90/30 precision in line with GS and TPDDTEC sampling principles, our sample size was 111. A minimum of 20 KS and 7 KPTs were conducted on households in project target areas. This provides a representative baseline of the whole project area.

Baseline Kitchen Survey results show the following household stove habits, averaged over both rainy and dry seasons.

Cooking Technology	Average number of uses of technology per day	Percentage of households using this technology
Three stone fire	2.7	52%
Kuni Mbili (Wood ICS)	2.5	48%
LPG Stove	1.1	33%

Three stone fires were found to be the primary stove in highland areas, compared to the Kuni Mbili which is more popular in lowland areas of the baseline study area. Highland areas are generally characterized by high cooking energy costs and lower temperatures. This means there is a need to heat homes, a specific design feature of the Wisdom Jiko. The primary focus of the project is these highland areas, replacing inefficient three stone fires. Wood ICS are also factored into the baseline and ex ante

calculations, as their use is widespread in lowland areas which are within the project boundary.

Regarding LPG Stove use, the baseline study process found that households who own LPG stoves use them on rarer occasions as a 'convenience device'. For example, those with LPG stoves will use them to reheat food but not to cook a meal from scratch. This is supported by the average number of uses of the technology per day. In addition, stove stacking is a factor to consider. Different stoves are present in households often for different purposes and to be used in different circumstances. The LPG stove thus falls outside of the scope of the project as its use will not be affected by the introduction of the improved project stove. Wisdom Energy Hub seeks to replace the use of inefficient wood fuel cooking technologies only. Project usage surveys will monitor this scenario.

The Kitchen Performance test was conducted on three stone fires in situ. 42 KPTs were completed over the course of 4 days. Results showed average wood usage of 10.59 kg/hh/day. Three stone fires are extremely inefficient and demand large quantities of fuelwood, resulting in high greenhouse gas emissions, deforestation and exposure to indoor air pollution, as shown by the KPTs. For the Kuni Mbili ICS, the conservative methodology default of 0.5 tonnes/capita/year has been chosen.

The baseline study also found the following information. A mix of collection and purchase of firewood is present. 48% of households only collect fuelwood, 29% buy and 23% do both. Of those that collect firewood 97% collect from close to farm fields.

The Baseline Kitchen Survey shows that households spend on average 58 minutes collecting firewood per trip, and take 1.85 trips per week, therefore, households spend an average of 1hr 49minutes a week collecting firewood a day. As the project stove reduces firewood consumption, the burden of collecting firewood will be significantly reduced.

On average, families spend 176 minutes per day cooking. This presents a high exposure to indoor air pollution caused by the smoke. The project stove reduces cooking time and smoke pollution. Therefore, the project will reduce cases of respiratory illnesses caused by the inhalation of smoke.

The project stoves will displace the inefficient baseline technology and reduce greenhouse gas emissions by reducing wood use by roughly 44%³. The reduced use and demand for firewood will reduce greenhouse gas emissions, reduce demand of wood collection on local forests (for those that collect wood), reduce requirement to purchase wood (for those who buy wood), reduce the time spent collecting wood and cooking, and reduce exposure to harmful indoor air pollution.

B.5. Demonstration of additionality

The table below is only applicable if the proposed project is deemed additional, as defined by the applied approved methodology or activity requirement or product requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	As demonstrated in the Gold Standard for the Global Goals Community Services Activity Requirements section 4.1.9 - Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B) (b) Projects located in LDC, SIDS, LLDC (c) Micro-scale projects
Describe how the proposed project meets the criteria for deemed additionality.	The project meets criterion 1. Positive list, Annex B of the Community Services Activity Requirements. Section 1.1.3 of Annex B states. “Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where

³ Weighted average of three stone fire and ICS baselines.

	each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year". Ex ante calculations have shown that emission reductions per improved cookstove per year come to 2.25 tCO₂e. The project is therefore additional.
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B.5.1. Prior Consideration

N/A

B.5.2. Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
SDG 3. – Good Health and Well Being	3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	HAPRy: total reduction in household air pollution for project activity in year y (%)
SDG 5. – Gender Equality	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	TRy: total reduction time spent collecting firewood and cooking for project activity in year y (%)

SDG 7. – Affordable and Clean Energy	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	ICS _{a,y} : The number of active ICS disseminated by the project, during year y
SDG 13. - Climate Action (mandatory)	13.B.1: Number of least developed countries and small island developing States with nationally determined contributions, long-term strategies, national adaptation plans, strategies as reported in adaptation communications and national communications	Emissions Reductions

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

SDG 3 (Good Health and Wellbeing)

Switching from an unimproved stove to an improved cookstove reduces the volume of non-renewable biomass that needs to be burned for the household's daily cooking needs. Therefore, the reduction in non-renewable biomass burned can be taken as a proxy indicator of how these VPAs contribute towards the SDG 3 target of substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.

The average difference in fuel use per technology per day in the project will be subtracted from the baseline figure which comes from the Kitchen Performance test and is fixed throughout the crediting period. This difference will then be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target.

The overall reduction in HAP by the project activity are then calculated as follows:

$$\text{HAPR}_y = ((\text{Pb},y - \text{Pp},y) / \text{Pb},y) * \text{Up},y$$

Where:

HAPR_y Total reduction in Household Air Pollution for project activity in year y (%)

Pb,_y Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)

P _{p,y}	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)
U _{p,y}	Usage rate in project scenario p during year y

SDG 5 (Gender Equality)

Women and girls perform the majority of unpaid domestic work⁴. This leaves them with less time to rest, study and realise their economic potential, leaving them in time poverty. In regard to time, women are poorer than men as unpaid domestic duties, such as collecting firewood and cooking, must be added to their market productive work, making time much scarcer⁵. Women are widely recognised as being principally responsible for natural resource collection⁶.

These trends demonstrate that reducing the amount of firewood required by households and access to a more efficient cooking technology has the potential to reduce the time poverty of women, because the time burden of collecting firewood and cooking, which falls disproportionately on women, will be reduced. The decrease per household in time spent gathering firewood and cooking will be taken as a proxy contribution towards the SDG target.

The overall reduction in time spent collecting firewood by the project activity are then calculated as follows:

$$TR_y = (T_{b,y} - T_{p,y}) / T_{b,y}$$

Where:

TR _y	Total reduction in time spent on cooking and firewood collection activities
T _{b,y}	Baseline time spent on cooking and firewood collection activities per household per day

⁴ UN (2017) 'Progress towards the Sustainable Development Goals (E/2017/66)'. Available at <https://unstats.un.org/sdgs/files/report/2017/secretary-general-sdg-report-2017--EN.pdf>

⁵ Charmes, J 'A Review of Empirical Evidence on Time Use in Africa from UN-Sponsored Surveys', in World Bank (2006) 'Gender, Times Use, and Poverty in Sub-Saharan Africa'. World Bank Working Paper No. 73

⁶ Nankhuni (2004) 'Environmental Degradation, Resource Scarcity and Children's Welfare in Malawi: School Attendance, School Progress, and Children's Health'

$T_{p,y}$ Project time spent on cooking and firewood collection activities per household per day

SDG 7 (Affordable and Clean Energy)

The VPA involves the distribution of ICS, a fuel-efficient technology, to replace the inefficient baseline cooking technologies.

To measure 7.1 (universal access to affordable, reliable and modern energy services), indicator 7.1.2 provides for the measurement of the 'proportion of the population with primary reliance on clean fuels and technology'. The outcome for SDG 7 is quantified as the number of active ICS disseminated by the project, during year y , and is calculated using parameters from the project scenario.

$$ICS_{a,y} = N_{p,y} * U_{p,y}$$

Where:

$ICS_{a,y}$ The number of active ICS disseminated by the project, during year y
 $N_{p,y}$ Project ICS distributed in year y
 $U_{p,y}$ Usage rate

SDG 13 (Climate Action)

The contribution to SDG 13 will be measured through carbon credits issued following the Gold Standard TPDDTEC v3.1 methodology.

Baseline Emissions

$$BE_{b,y} = B_{b,y} * ((f_{NRB} * EF_{b,fuel, CO2}) + EF_{b,fuel, nonCO2}) * NCV_{b, fuel}$$

Where:

$BE_{b,y}$ Emissions for baseline scenario b during the year y in tCO_2e
 $B_{b,y}$ Quantity of fuel consumed in baseline scenario b during year y , in tons, as per by-default factors (cases with project performance field test only)
 f_{NRB} Fraction of biomass used during year y for the considered scenario that can be established as non-renewable biomass
 $NCV_{b,fuel}$ Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)

EF_{b,fuel,CO2} CO₂ emission factor of the fuel that is substituted or reduced. 112 tCO₂/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel

EF_{b,fuel,nonCO2} Non-CO₂ emission factor of the fuel that is substituted or reduced

$$B_{b,y} = N_{p,y} * P_{b,y}$$

Where:

N_{p,y} Project technology-days in the project database for project scenario p through year y

P_{b,y} Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day

Project Emissions

$$PE_{p,y} = B_{p,y} * ((f_{NRB} * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p,fuel}$$

Where:

PE_{p,y} Emissions for project scenario p during year y in tCO₂e

B_{p,y} Quantity of fuel consumed in project scenario p during year y, in tons

f_{NRB} Fraction of biomass used during year y that can be established as non-renewable biomass

NCV_{p,fuel} Net calorific value of the project fuel (IPCC default for wood fuel, 0.015 TJ/ton). This is equal to the baseline fuel NCV in projects which use the same fuel.

Emission Reduction Calculation

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y}$$

Where:

ER_y Emission reduction for total project activity in year y (tCO₂e/yr)

BE_{p,y} Baseline emissions for baseline scenario b in year y (tCO₂e/yr)

PE_{b,y} Project emissions for project scenario p in year y (tCO₂e/yr)

LE_{p,y} Leakage for project scenario p in year y (tCO₂e/yr)

B.6.2. Data and parameters fixed ex ante

SDG 13

Data/parameter	$EF_{b,fuel,CO_2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of wood-fuel in baseline scenario (includes production, transport and use)
Source of data	Calculated from IPCC defaults Volume 2: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of baseline emissions.
Additional comment	
	-

Data/parameter	$EF_{b,non\ CO_2}$			
Unit	tCO ₂ e/TJ			
Description	Non-CO ₂ (CH ₄ and N ₂ O) emission factor arising from use of wood fuel in baseline scenario			
Source of data	IPCC Default emissions factor			
Value(s) applied	9.460			
Choice of data or Measurement methods and procedures	Deemed valid by Methodology			
		Default Emissions factor (kg_gas/TJ _{NCV})	GWP of gas	Default Emissions factor (kg_CO ₂ e/TJ _{NCV})
	Gas			Default Emissions factor (t_CO ₂ e/TJ _{NCV})
	CH ₄	300	28	8,400

	<table><tr><td>N₂O</td><td>4</td><td>265</td><td>1,060</td><td>1.060</td></tr><tr><td></td><td></td><td></td><td>Total</td><td>9.460</td></tr></table>	N ₂ O	4	265	1,060	1.060				Total	9.460
N ₂ O	4	265	1,060	1.060							
			Total	9.460							
Purpose of data	Calculation of emission reductions.										
Additional comment	-										

Data/parameter	EF _{p,co2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of wood fuel in project scenario
Source of data	Volume 2: 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	EF _{p,non co2}				
Unit	tCO2e/TJ				
Description	Non-CO2 (CH4 and N2O) emission factor arising from use of wood fuel in project scenario				
Source of data	IPCC Default emissions factor				
Value(s) applied	9.460				
Choice of data or Measurement methods and procedures	Deemed valid by Methodology				
		Default Emissions factor (kg_gas/TJ _{NCV})	GWP of gas	Default Emissions factor (kg_CO2e/TJ _{NCV})	Default Emissions factor (t_CO2e/TJ _{NCV})
	Gas				
	CH4	300	28	8,400	8.4000
	N2O	4	265	1,060	1.060

				Total	9.460
Purpose of data	Calculation of emission reductions.				
Additional comment	-				

Data/parameter	NCV _{fuel}
Unit	TJ/ton
Description	Net calorific value of the wood fuel used in the baseline
Source of data	IPCC Default emissions factor
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	Weighting _{3SF}
Unit	%
Description	Percentage of baseline scenario that is comprised of three stone fire (3sf) cooking technology
Source of data	Baseline Kitchen Survey
Value(s) applied	52%
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculation of emission reductions.

Additional comment	-
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Data/parameter	Weighting _{ICS}
Unit	%
Description	Percentage of baseline scenario that is comprised of three Improved Cookstove (ICS) technology
Source of data	Baseline Kitchen Survey
Value(s) applied	48%
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Data/parameter	P _{b,y}
Unit	kg/household/day
Description	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)
Source of data	Baseline Kitchen Performance Test
Value(s) applied	9.03
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculating quantity of fuel used in the baseline scenario.
Additional comment	Weighted average of three stone fire and Improved Woodfuel Stove identified in baseline scenario. Using parameters Weighting _{3SF} and Weighting _{ICS} .

SDG 5

Data/parameter	$T_{b,y}$
Unit	Hours
Description	Time spent on cooking and firewood collection activities per household per week prior to project in year y
Source of data	Baseline Kitchen Survey questions 28, 33, 35
Value(s) applied	3.19
Choice of data or Measurement methods and procedures	Measured by question in the baseline kitchen survey.
Purpose of data	Calculating time saved collecting firewood and cooking by project.
Additional comment	-

B.6.3. Ex ante estimation of SDG Impact

SDG 3: Good Health and Well Being	Description	$HARPy = ((P_{b,y} - P_{p,y})/P_{b,y}) * Up,y$	Unit
HARPy	Total reduction in Household Air Pollution for project activity in year y	59%	%
$P_{b,y}$	Quantity of fuel that is consumed in the baseline scenario b during year y	9.03	kg/hh/day
$P_{p,y}$	Quantity of fuel that is consumed in the project scenario p during year y	2.36	kg/hh/day
Up,y	Usage rate in project scenario p during year y	80%	%

SDG 5: Gender Equality	Description	$TRy = (T_{b,y} - T_{p,y}) / T_{b,y}$	Unit
TRy	Total reduction time spent on cooking and firewood collection activities	16%	%
$T_{b,y}$	Baseline time spent on cooking and firewood collection activities per household per day	3.19	hours
$T_{p,y}$	Project time spent on cooking and firewood collection activities per household per day	2.69	hours

SDG 7: Affordable and Clean Energy	Description	$N_{p,y} * Up,y$	Unit
Active ICS disseminated by project	The number of active ICS disseminated by the project, during year y	2,400	Number
$N_{p,y}$	Number of stoves distributed, during year y	3,000	Number
Up,y	Usage rate in project scenario during year y	80%	%

SDG 13: Climate Action	Description		Unit
Er,y	CO2 emission reductions for the annual average emission reductions	23,863	VER

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 3

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0% reduction in exposure to Household Air Pollution	59% reduction in exposure to Household Air Pollution	59% reduction in exposure to Household Air Pollution
Year 2	0%	59%	59%
Year 3	0%	59%	59%
Year 4	0%	59%	59%
Year 5	0%	59%	59%
Total	0%	59%	59%
Total number of crediting years	5		
Annual average over the crediting period	0% reduction in exposure to Household Air Pollution	59% reduction in exposure to Household Air Pollution	59% reduction in exposure to Household Air Pollution

SDG 5

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0% reduction in time spent cooking and collecting firewood	16% reduction in time spent cooking and collecting firewood	16% reduction in time spent cooking and collecting firewood
Year 2	0%	16%	16%
Year 3	0%	16%	16%
Year 4	0%	16%	16%
Year 5	0%	16%	16%
Total	0%	16%	16%
Total number of crediting years	5		
Annual average over the crediting period	0% reduction in time spent cooking and collecting firewood	16% reduction in time spent cooking and collecting firewood	16% reduction in time spent cooking and collecting firewood

SDG 7

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0 active ICS	2,400	2,400
Year 2	0	2,400	2,400
Year 3	0	2,400	2,400
Year 4	0	2,400	2,400
Year 5	0	2,400	2,400
Total	0	12,000	12,000
Total number of crediting years	5		
Annual average over the crediting period	0 active ICS	2,400 active ICS	2,400 Active ICS

SDG 13

Year ⁷	Baseline estimate	Project estimate	Net benefit
Year 1	11,685 tCO ₂ e	7,346 tCO ₂ e	4,339 tCO ₂ e
Year 2	29,212	18,365	10,847
Year 3	46,739	29,384	17,355
Year 4	64,266	40,403	23,863
Year 5	81,793	51,422	30,371
Year 6	87,636	55,095	32,540
Total	321,330	202,016	119,314
Total number of crediting years	5		

⁷ The Crediting Period is from 01/05/2021 – 30/04/2026. Years 1 and 6 are therefore not full 365-day periods.

Annual average over the crediting period	64,266 tCO ₂ e	40,403 tCO ₂ e	23,863 tCO₂e
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B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	N _{p,y}
Unit	Project Technology Days
Description	Technologies in the project database for project scenario p through year
Source of data	Cookstove Database
Value(s) applied	To be monitored
Measurement methods and procedures	Sum of the total number of cookstoves distributed multiplied by the number of days crediting each technology earns in a given monitoring period
Monitoring frequency	Continuous recording
QA/QC procedures	Sales records are analysed prior to calculation of parameter
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter	U _{p,y}
Unit	Percentage
Description	Usage rate in project scenario p through year y
Source of data	Usage Survey
Value(s) applied	Estimated at 80%. To be monitored
Measurement methods and procedures	Usage Survey
Monitoring frequency	Measured annually
QA/QC procedures	The usage survey will be carried out by trained local staff to meet the specific requirements of the methodology. All

	data presented in excel is subject to checking and cross referencing of a sample of the raw data by CO2balance
Purpose of data	Calculation of usage
Additional comment	-

Data / Parameter	$f_{NRB,i,y}$
Unit	Fractional non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Calculated in line with CDM Tool 30 EB 108 Annex 11 v3.0 2020
Value(s) applied	0.93
Measurement methods and procedures	CDM Tool 30 EB 108 Annex 11 v3.0 2020
Monitoring frequency	Annual
QA/QC procedures	
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter	$LE_{p,y}$
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Monitoring Kitchen Surveys supported by desk-based research (leakage assessment)
Value(s) applied	Estimated at 0. To be monitored
Measurement methods and procedures	Leakage Assessment
Monitoring frequency	Leakage factors are re-assessed biennially.
QA/QC procedures	Research and surveys checked by senior project staff.
Purpose of data	Calculation of leakage.
Additional comment	-

Data / Parameter	Pp,y
Unit	Kg/household
Description	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)
Source of data	Kitchen Performance Test
Value(s) applied	To be monitored
Measurement methods and procedures	Calculating quantity of fuel used in the project scenario using the KPT survey
Monitoring frequency	Measured biennially
QA/QC procedures	Training to be given to survey conductors by CO2balance. Measurement scales to be calibrated biennially, with first calibration completed prior to first verification.
Purpose of data	Calculation of SDG 3
Additional comment	Should more than one fuel type be present and significant (i.e. greater than 5% of project survey respondents using;) in the project scenario, Pp,y shall be split per fuel type

SDG 3

Data / Parameter	HARP,y
Unit	percentage
Description	Total reduction in Household Air Pollution for project activity in year y (%)
Source of data	Project kitchen Survey and baseline kitchen survey
Value(s) applied	To be monitored
Measurement methods and procedures	The average difference in fuel use per technology per day in the project will be subtracted from the baseline figure. This difference will then be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target
Monitoring frequency	Measured annually
QA/QC procedures	

Purpose of data	To calculate the overall reduction in household air pollution from the project activity compared to the baseline scenario
Additional comment	This is parameter is used to monitor safe guarding principle 3: Community Health, Safety and Working Conditions, as per Section D.1.

SDG 5

Data / Parameter	TP,y
Unit	Hours
Description	Time spent collecting firewood and cooking per household per week in the project in year y
Source of data	Monitoring Kitchen Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Assessed every year using Project Survey
Monitoring frequency	Measured annually
QA/QC procedures	Training to be given to survey conductors by CO2balance
Purpose of data	Calculation of SDG 5
Additional comment	-

Data / Parameter	TR,y
Unit	hours
Description	Total reduction in time spent collecting firewood and cooking per household in the project activity in year y
Source of data	Project Survey and baseline survey.
Value(s) applied	To be monitored
Measurement methods and procedures	Assessed every year using Project Survey. Calculate the average amount of time spent collecting firewood in the project scenario and compare to the pre-project scenario
Monitoring frequency	Measured annually
QA/QC procedures	

Purpose of data	To quantify whether the project has contributed to a reduction in the amount of time spent collecting firewood compared to the pre-project scenario towards SDG 5 target.
Additional comment	-

SDG 7

Data / Parameter	$N_{p,y}$
Unit	Number of stoves
Description	Number of stoves distributed during year y
Source of data	Sales Record
Value(s) applied	Estimated at 3,000. To be monitored
Measurement methods and procedures	The Sales Record of the Project will be updated at least annually.
Monitoring frequency	Annually
QA/QC procedures	CO2balance will request supporting evidence of sales record from VPA Implementer, Wisdom Energy Hub Ltd
Purpose of data	To determine number of Active ICS disseminated by project
Additional comment	-

Safeguarding Principle: Corruption

Data / Parameter	Reported cases of corruption arising from project activity
Unit	
Description	Report on any cases of corruption arising from the project activity communicated by communities
Source of data	Continuous input mechanism
Value(s) applied	
Measurement methods and procedures	The communities will be able to communicate any cases of corruption through the continuous input mechanism. The continuous input mechanism will be monitored and any reports of corruption will be acted on.
Monitoring frequency	Annually

QA/QC procedures	Monitor continuous input mechanism and review project information regularly.
Purpose of data	To monitor 'Corruption' safeguarding principle.
Additional comment	-

B.7.2. Sampling plan

Cross sampling of devices will be applied across all homogenous VPAs in the project area. Homogenous VPAs are defined as those that are sharing a common baseline. The number of districts in the project area that will need to be sampled for a 90/30 confidence/precision will be determined. From the determined number of districts, a village from each district will be randomly selected and project households will be randomly sampled, complying with the minimum sample size for the particular survey/test.

Individual participants will be randomly selected from the cookstove electronic database. Sample sizes will be in line with the Gold Standard requirements. The random sample group is reselected for every monitoring period to ensure the selection remains random.

The surveys below will be monitored under the cross-sampling approach:

- Project Surveys- Completed annually
- Usage Surveys- Completed annually
- Kitchen Performance Test- Completed biennially

The surveys will be conducted so as to ensure that they are within the end date of the respective monitoring periods for each VPA.

B.7.3. Other elements of monitoring plan

This VPA, and any future homogenous VPAs, will be filled with improved stoves sold under the project, each with a unique identification number imprinted to the stove. Purchasers of the stove will be informed about the carbon transfer with the terms set out in the warranty receipt.

Sales Record

During the sales process, representatives of Wisdom Energy Hub will collect data and educate recipients on proper usage of the technology.

A comprehensive Sales Record will record the following information and will be backed up electronically. The representatives will endeavor to collect all of the following information in order to build a comprehensive Sales Record. Section 3.1.A page 29 (footnote 31) of TPDDTEC v3.1 provides that details of end users are not required in cases it is "*not feasible*" to collect them, in particular cases where "*sales of small items (such as portable cook stoves, heat retention cookers, water filters) sold in market stalls or shops where the retailer cannot reasonably be expected to collect customers names and addresses during busy times*". Due to the nature of stove dissemination in this Project, this caveat may be relied upon where it is not feasible for the vendor to collect all information. Due to the volume of sales expected (3000 per year) and the training of LMDs, the Sales Record satisfy the size requirement to "*ensure an adequate end user pool to which random sampling can be applied*". It is in the interest of the vendor to collect the end user information, including telephone number, as each user receives follow up.

The following data will be collected:

- Date of sale
- Geographic area of sale
- Model/type of project technology sold
- Quantity of project technologies sold
- Name, address and telephone number of end user (where feasible)
- Unique stove ID

Carbon Transfer Form

The vendee will be informed about the transfer of carbon rights at the point of sale. Each vendee receives a warranty receipt which again states the terms of the transfer of carbon rights.

The details will be added to the electronic database which will provide the value for PTDs and be used to randomly select the monitoring samples.

Project Database

The Project Database is derived from the Sales Record. The Database tracks the active stoves which provides the value for PTDs and is used to randomly select the monitoring samples.

Ongoing Monitoring Studies

All monitoring studies will be recorded on a hard copy which the monitoring team will transfer into digital format for analysis documentation and reporting by Wisdom Energy Hub. The hard copy is scanned and then filed to enable transparent auditing.

a) Monitoring Kitchen Survey

- Completed annually.
- Begins within 1 year after project registration.
- Surveys end users currently using project technologies to explore changes in project scenario over time.
 - The survey will monitor fuel use in the project scenario. Should different fuel uses be present and significant (greater than 5% of results), this information will feed into KPTs conducted. More information is presented under c).
- Sample size guidance same as Baseline Survey:
 - Group Size <300: Minimum sample size 30 or population size, whichever is smaller.
 - Group Size 300 – 1000: Minimum sample size 10% of group size
 - Group Size > 1000: Minimum sample size 100

b) Usage Survey

- Completed annually
- Provides a usage rate

Usage Surveys will be carried out in line with the Good Practice Monitoring Requirements included in the Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices Rule Update, including:

- Kitchen observation – usage surveys will be carried out in each household in order to gather objective information.
- Interview with the primary cook – each usage survey interview will be carried out to gather information on stove use patterns.
- Photos of the cooking area(s) – a photograph will be taken of the cooking area at the end of each usage survey upon consent of the interviewee
- GPS co-ordinates – GPS co-ordinates will be gathered during each usage survey
- Field team training and supervision – training will be conducted annually with the field team before the surveys are carried out. Data will be reviewed regularly as it is collected to ensure that the expected standard was maintained
- End-User Training, follow up visits and Awareness Campaign – demonstrations and training take place at the point of sale. Where feasible, all sales are followed up with a phone call to collect the users' feedback

and advise of stove usage. Awareness Campaigns will encourage end users to discontinue their baseline stove use.

- Verification checks - 5-10% of surveys will be verified by phone after the monitoring has been conducted

In line with the Good Practice Monitoring Requirements the usage rate will be capped at 90% in each monitoring period.

c) Project FT Update

- Completed biennially and in time for first verification.
- Extension of the KPT to provide fuel consumption representation of project technologies currently in use.
 - KPTs will always be based on wood fuel. In addition, should the monitoring kitchen survey results show different significant fuel uses (i.e., greater than 5% of respondents using), then KPTs will be extended to cover those fuels.
- Carried out by Wisdom Energy Hub under the supervision of CO2balance.
- Guidance on the sampling strategy will be provided by CO2balance.

d) Leakage Assessment

- Completed biennially.
- The potential sources of leakage will be investigated. If the assessment quantifies an increase in fuel consumption by the non-project households attributable to the project activity, then calculations will be adjusted to account for this.

e) Project Technology Days

- The number of project technology days is a measure of the number of technologies sold and the number of days each stove has been operating within a crediting period.
- The total number of project technology days will be calculated by using the date of each individual stove installation completion, indicated by the date of signed Carbon Transfer Form, to determine the number of days it operated within a crediting period. The profile for each individual stove within the project will then be summed to derive the total number of project technology days.

DOE Validation and Verification Site Visits

An independent Designated Operational Entity (DOE) will be appointed to review the Project and conduct a site visit as part of the Validation and Verification processes. As this VPA forms part of a group of homogenous VPAs, a DOE site visit at Validation and Verification is only required for the first VPA in the group. Subsequent homogenous VPAs that are Included or Verified under the PoA will undergo a Compliance Check to ensure that they meet the criteria provided in the PoA-DD.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

30/04/2021

C.1.2. Expected operational lifetime of project

15 years (5 years twice renewable)

C.2. Crediting period of project

C.2.1. Start date of crediting period

01/05/2021

C.2.2. Total length of crediting period

5 years twice renewable

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3. Community Health, Safety and Working Conditions	Will be measured through SDG 3
Principle 5. Corruption	The communities will be able to communicate any cases of corruption through the continuous input mechanism. The continuous input mechanism will be monitored and any reports of corruption will be acted on.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project will reduce the amount of resources required to cook. There will be less burden on women, men and children, as less firewood for cooking needs to be collected. This will reduce the time burden on women and men in the socially isolating activity of collecting resources. Women are encouraged to attend the LSC, and have special platforms to express their views and concerns of the Project. Women and men will have more time to engage in activities due to the reduced time required to be spent gathering firewood. Both women and men will benefit from the project activities, no group is excluded from participating in the project activities or use of the stoves. The project will decrease the workload of women in collecting firewood and time spent cooking, thereby allowing more time to engage in other activities. The project aims to benefit the whole community equally and women's equal participation in the LSC is encouraged. The Project will increase natural protection as no firewood will be required for cooking. No further risks or hazards for women were identified.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	Official government documentation on Gender Policy in Kenya consists of the National Gender and Equality Commission Strategic Plan for the period 2019 – 2024 and National Policy on Gender and Development. These policy documents will be used as a guide to align the project with current national gender policies and best practices in Kenya. Some of the main objectives of these policies and how the project aligns with these are as follows:

Objective 1: "To enhance awareness on the principles of equality and inclusion"⁸

The project seeks to promote gender equality in all project related activities, in particular the majority of sales are done through demonstrations at women's groups, meaning that women are at the for front of the decision to buy an improved Cook Stove. . The project aims to impact upon sustainable development within the communities it touches, including contributing to the sustainable development goals, one of which is SDG 5 gender equality. The impact of the project on SDG 5 will be measured on an annual basis during project monitoring to determine the contribution towards the SDG 5 target 5.4 - Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate.

"Ensure women participate in and benefit equitably from investments in various natural resources development initiatives"⁹

This project to this policy objective as the project is an initiative that reduces demand on natural resources. The Safe improved cookstoves will reduce the burden on both women and men of to collect firewood, additionally women in particular benefit from reduce time cooking and exposure to smoke. This gives them more time and opportunity to pursue development enhancing activities such as income generating activities and education, which may have once been sacrificed to provide firewood for the

⁸ Strategic Plan 2019-2024, National Gender and Equality Commission. 2019
<http://www.ngeckenya.org/Downloads/NGEC-Strategic-Plan-2019-2024.pdf>

⁹ National Policy on Gender and Development, Sessional Paper No. 02 of 2019, The State Department of Gender Affairs. 2019. <https://gender.go.ke/wp-content/uploads/2021/04/Final-NPGAD-March-2021.pdf>

	homestead. This acts to close the gender gap by allowing women and girls more time for self-improvement and development. The project survey will monitor the impacts of time saved from the project and what activities respondents are choosing to do with their time saved. This helps beneficiaries achieve a more sustainable livelihood over the project lifetime.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

Stakeholder consultation has not been conducted yet, due to Covid-19 postponement.

E.2. Final continuous input / grievance mechanism

	Method Chosen (include all known details e.g. location of book, phone, number, identity of mediator)	Justification of Choice (best practice)
Continuous Input / Grievance Expression Process Book (mandatory)	Wisdom Energy Hub and CO2balance contact information is provided on stove receipts at sale. End users can contact either party. Input/Grievances are logged accordingly	Project utilises a sales model, it is more suitable to give flexible feedback options than a book held in location.

GS Contact (mandatory) help@goldstandard.org

Telephone access (optional)	+254 (0) 700 011 365
Internet/email access (optional)	projects@co2balance.com sales@wisdomenergy.co.ke
Nominated Independent Mediator (optional)	
Other	

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project will adhere to all human rights requirements including respecting internationally proclaimed human rights and Universal Declaration of Human Rights and will not discriminate in any way.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead	1. No	Both women and men will benefit from the project	

to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	2. No 3. No 4. No	activities, no group is excluded from participating in the project activities or use of the stoves. The project decreases the workload of women in collecting firewood and cooking, thereby allowing more time to engage in other activities. No further risks or hazards for women and girls have been identified.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes	The project will reduce the community exposure to illnesses associated with household air pollution through the distribution of improved cookstoves	

Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area does not include cultural and historic sites. The focus of the project is distributing improved cookstoves only.	
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The focus of the project is distributing improved cookstoves only.	
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	No changes to land tenure arrangements and/or rights are required.	
>>			
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	Yes	The project shall ensure that all forms of corruption are avoided. Project beneficiaries are able to contact the project developer and implementer through the continuous grievance mechanism to report any form of corruption.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include:	No	The project adheres to all labour laws and requirements	

a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)			
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5.	The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences				
1.	Does the project cause negative economic consequences during and after project implementation?	No	There are no negative economic impacts and potential risks to local economy.	
>>				
Principle 7.1 Emissions				
	Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	GHG emissions will be reduced through replacing less efficient cooking methods with improved cook stoves.	
>>				
Principle 7.2 Energy Supply				
	Will the Project use energy from a local grid or power	No	The project will reduce consumption of biomass.	

supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?			
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	There will be no significant change in the volume of water consumed by the households.	
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The focus of the project is distributing improved cookstoves only	
>>			
Principle 9.1 Landscape Modification and Soil			

Does the Project involve the use of land and soil for production of crops or other products?	No	No crops or other products are produced in the project.	
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	There will be no impact by the project to natural disasters.	
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	No GMOs will be used in the project.	
>>			

Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The Project reduces the amount of pollutants released into the air.	
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project does not deal with hazardous or non-hazardous chemicals and/or materials.	
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	No pesticides and/or fertilisers are to be used in the project.	
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project reduces the consumption of firewood, therefore having a positive impact on forest conservation.	
>>			
Principle 9.8 Food			

Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project has no impact on modifying foods or altering crop regimes.	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project does not involve animal husbandry.	
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project decreases the consumption of firewood having a positive impact on conserving forest ecosystems.	
>>			
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including	No	There are several endangered species in Kenya. The project is not envisaged to have any impact on their habitat as it	

those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		only affects solar desalination plant infrastructure. The project does not impact other areas where endangered species are present.	
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APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	CO2balance UK Ltd
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APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of [Principles and Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption